Brutal Series - Paper B17 (2x): Nutrition in Animals / Exponents & Powers / Transportation in Animals & Plants / Perimeter & Area

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. A meal holds 80 g of starch. In the mouth, salivary amylase breaks down 15% of it. Of the starch still left, the small intestine digests three-quarters. How many grams of starch stay undigested? [Nutrition in Animals]

- **(A)** 17 g
- **(B)** 12 g
- **(C)** 11 g
- **(D)** 20 g
- **(E)** 60 g
- **(F)** 51 g
- **(G)** 68 g

2. Simplify and find the value: (2^3 x 2^4) / 2^5. [Exponents & Powers]

- **(A)** 32
- **(B)** 2
- **(C)** 4
- **(D)** 16
- **(E)** 2
- **(F)** 8
- **(G)** 64

3. A person's heart beats 72 times each minute while resting. How many times does it beat in one full hour? [Transportation in Animals & Plants]

- **(A)** 4320
- **(B)** 1440
- **(C)** 7200
- **(D)** 72
- **(E)** 8640
- **(F)** 720

4. A rectangular field is 25 m long and 16 m wide. Fencing wire costs Rs 8 per metre. What is the cost to fence all four sides once? [Perimeter & Area]

- **(A)** Rs 656
- **(B)** Rs 3200
- **(C)** Rs 800
- **(D)** Rs 41
- **(E)** Rs 1312
- **(F)** Rs 328
- **(G)** Rs 82

5. A cow eats 18 kg of grass. It quickly swallows five-sixths of it into the rumen and chews the rest at once. Later it brings back one-third of the rumen grass as cud. How much cud does it chew later? [Nutrition in Animals]

- **(A)** 5 kg
- **(B)** 9 kg
- **(C)** 6 kg
- **(D)** 13 kg
- **(E)** 10 kg
- **(F)** 3 kg
- **(G)** 15 kg

6. Find the value of (3^2)^3 / 3^4. [Exponents & Powers]

- **(A)** 81
- **(B)** 3
- **(C)** 243
- **(D)** 18
- **(E)** 6
- **(F)** 9
- **(G)** 27

7. A nurse counts 18 pulse beats in 15 seconds at the wrist. What is the pulse rate per minute? [Transportation in Animals & Plants]

- **(A)** 90
- **(B)** 108
- **(C)** 270
- **(D)** 60
- **(E)** 72
- **(F)** 36

8. A square has a perimeter of 48 cm. What is its area? [Perimeter & Area]

- **(A)** 2304 cm^2
- **(B)** 96 cm^2
- **(C)** 12 cm^2
- **(D)** 144 cm^2
- **(E)** 48 cm^2
- **(F)** 576 cm^2
- **(G)** 36 cm^2

9. An amoeba feeds by pushing out finger-like extensions that flow around a food particle and trap it inside a food vacuole. What is this feeding method called? [Nutrition in Animals]

- **(A)** Filtering food through tiny cilia
- **(B)** Engulfing food using pseudopodia (false feet)
- **(C)** Absorbing dissolved food through roots
- **(D)** Biting and grinding food with teeth
- **(E)** Trapping prey with stinging tentacles
- **(F)** Sucking liquid food with a proboscis
- **(G)** Sieving food particles with gills

10. Write the number 47,000 in standard (scientific) form. [Exponents & Powers]

- **(A)** 470 x 10^2
- **(B)** 4.7 x 10^4
- **(C)** 4.7 x 10^3
- **(D)** 4.7 x 10^2
- **(E)** 0.47 x 10^5
- **(F)** 47 x 10^3

11. Blood is red because one type of cell carries oxygen using a red pigment. Which cells are these? [Transportation in Animals & Plants]

- **(A)** Plasma, the liquid part of blood
- **(B)** Nerve cells, which carry signals
- **(C)** Red blood cells, carrying oxygen with haemoglobin
- **(D)** White blood cells, which fight germs
- **(E)** Bone cells, which store calcium
- **(F)** Platelets, which clot the blood
- **(G)** Skin cells, which protect the body

12. A triangle has a base of 14 cm and a height of 9 cm. What is its area? [Perimeter & Area]

- **(A)** 45 cm^2
- **(B)** 63 cm^2
- **(C)** 31.5 cm^2
- **(D)** 126 cm^2
- **(E)** 112 cm^2
- **(F)** 49 cm^2
- **(G)** 23 cm^2

13. In a human the small intestine is about 7.5 m long and the large intestine about 1.5 m. What is the total length of these two parts in centimetres? [Nutrition in Animals]

- **(A)** 9 cm
- **(B)** 150 cm
- **(C)** 9000 cm
- **(D)** 900 cm
- **(E)** 90 cm
- **(F)** 600 cm
- **(G)** 750 cm

14. Find the value of (-2)^3 + (-2)^2. [Exponents & Powers]

- **(A)** 0
- **(B)** -12
- **(C)** 4
- **(D)** -4
- **(E)** -2
- **(F)** 12

15. When a thorn pricks the skin and germs get in, which blood cells rush to fight the infection? [Transportation in Animals & Plants]

- **(A)** Bone marrow only
- **(B)** Red blood cells
- **(C)** Nerve cells
- **(D)** White blood cells
- **(E)** Plasma cells of fat
- **(F)** Platelets
- **(G)** Muscle cells

16. A parallelogram has a base of 12 cm and a height of 7 cm. What is its area? [Perimeter & Area]

- **(A)** 42 cm^2
- **(B)** 84 cm^2
- **(C)** 19 cm^2
- **(D)** 96 cm^2
- **(E)** 21 cm^2
- **(F)** 168 cm^2
- **(G)** 38 cm^2

17. An adult mouth has 32 teeth: 8 incisors, 4 canines, 8 premolars and 12 molars. What fraction of the teeth are the grinding teeth (premolars and molars)? [Nutrition in Animals]

- **(A)** 3/8
- **(B)** 3/4
- **(C)** 5/8
- **(D)** 5/16
- **(E)** 1/4
- **(F)** 2/3
- **(G)** 1/8

18. Find the value of (-1)^15 + (-1)^20. [Exponents & Powers]

- **(A)** 30
- **(B)** 35
- **(C)** -2
- **(D)** 2
- **(E)** 0
- **(F)** -1
- **(G)** 1

19. Which statement about blood vessels is correct? [Transportation in Animals & Plants]

- **(A)** Arteries have valves but veins do not
- **(B)** Arteries carry blood away from the heart at high pressure
- **(C)** Arteries always carry dirty blood
- **(D)** Veins have the thickest walls of all
- **(E)** Capillaries carry blood only to the heart
- **(F)** Capillaries are the thickest vessels

20. A circle has a radius of 7 cm. Taking pi = 22/7, what is its circumference? [Perimeter & Area]

- **(A)** 88 cm
- **(B)** 22 cm
- **(C)** 11 cm
- **(D)** 49 cm
- **(E)** 44 cm
- **(F)** 14 cm
- **(G)** 154 cm

21. A meal contains 24 g of fat. Bile from the liver breaks the fat into tiny drops so that seven-eighths of it is absorbed in the small intestine. How many grams of fat are NOT absorbed? [Nutrition in Animals]

- **(A)** 6 g
- **(B)** 2 g
- **(C)** 3 g
- **(D)** 21 g
- **(E)** 7 g
- **(F)** 18 g
- **(G)** 8 g

22. Find the value of (5^0 + 3^0) x 2^3. [Exponents & Powers]

- **(A)** 64
- **(B)** 0
- **(C)** 8
- **(D)** 16
- **(E)** 2
- **(F)** 4
- **(G)** 10

23. Water rises from the roots to the top leaves of a very tall tree mainly because of which pulling force? [Transportation in Animals & Plants]

- **(A)** Sugar pushing water down the phloem
- **(B)** Roots squeezing water like a sponge once a year
- **(C)** Wind pushing water up the trunk
- **(D)** Transpiration pull caused by water evaporating from the leaves
- **(E)** Sunlight melting the water upward
- **(F)** Gravity pulling the water upward

24. Using pi = 22/7, find the area of a circular plate whose radius is 7 cm. [Perimeter & Area]

- **(A)** 14 cm^2
- **(B)** 98 cm^2
- **(C)** 308 cm^2
- **(D)** 49 cm^2
- **(E)** 44 cm^2
- **(F)** 154 cm^2
- **(G)** 22 cm^2

25. The salivary glands make about 1500 mL of saliva each day. Over one full week, how many litres of saliva is that? [Nutrition in Animals]

- **(A)** 7.5 L
- **(B)** 1.05 L
- **(C)** 15 L
- **(D)** 10500 L
- **(E)** 1.5 L
- **(F)** 105 L
- **(G)** 10.5 L

26. By how much is 2^5 greater than 5^2? [Exponents & Powers]

- **(A)** 2
- **(B)** 10
- **(C)** 0
- **(D)** 3
- **(E)** 7
- **(F)** 57
- **(G)** 13

27. In a plant, two kinds of tubes carry materials. Which pair correctly states what each carries? [Transportation in Animals & Plants]

- **(A)** Both carry only food
- **(B)** Xylem carries water upward; phloem carries food made in leaves
- **(C)** Phloem carries water up; xylem carries food down
- **(D)** Xylem carries food; phloem carries water
- **(E)** Both xylem and phloem carry only water
- **(F)** Xylem carries oxygen; phloem carries water
- **(G)** Xylem carries blood; phloem carries air

28. A rectangular sheet measures 10 cm by 6 cm. A small square of side 2 cm is cut out from one corner. What is the area of the sheet that is left? [Perimeter & Area]

- **(A)** 56 cm^2
- **(B)** 60 cm^2
- **(C)** 64 cm^2
- **(D)** 52 cm^2
- **(E)** 58 cm^2
- **(F)** 48 cm^2

29. Of 500 g of food, the small intestine absorbs 80% as nutrients. From the remaining material, the large intestine absorbs 70% as water. What mass leaves the body as faeces? [Nutrition in Animals]

- **(A)** 400 g
- **(B)** 30 g
- **(C)** 350 g
- **(D)** 130 g
- **(E)** 70 g
- **(F)** 100 g

30. Express 1024 as a power of 2. [Exponents & Powers]

- **(A)** 10^2
- **(B)** 4^10
- **(C)** 2^12
- **(D)** 2^10
- **(E)** 2^8
- **(F)** 2^11
- **(G)** 2^9

31. Healthy kidneys make about 1.5 L of urine each day. How many millilitres of urine is made over 2 days? [Transportation in Animals & Plants]

- **(A)** 750 mL
- **(B)** 1.5 mL
- **(C)** 3 mL
- **(D)** 3000 mL
- **(E)** 1500 mL
- **(F)** 300 mL

32. A floor is 4 m long and 3 m wide. It is to be covered with square tiles each measuring 50 cm by 50 cm. How many tiles are needed? [Perimeter & Area]

- **(A)** 60
- **(B)** 96
- **(C)** 144
- **(D)** 12
- **(E)** 6
- **(F)** 48
- **(G)** 24

33. A cow's stomach has four chambers. If the rumen alone holds 60% of the stomach's total 200 L capacity, how many litres does the rumen hold? [Nutrition in Animals]

- **(A)** 120 L
- **(B)** 20 L
- **(C)** 200 L
- **(D)** 100 L
- **(E)** 80 L
- **(F)** 60 L
- **(G)** 140 L

34. Simplify 2^3 x 3^3 into a single power, then give its value. [Exponents & Powers]

- **(A)** 36
- **(B)** 216
- **(C)** 6^9
- **(D)** 729
- **(E)** 18
- **(F)** 6^6
- **(G)** 30

35. The heart has four chambers. Which two chambers contract to push blood OUT of the heart into the arteries? [Transportation in Animals & Plants]

- **(A)** All four chambers at once
- **(B)** One atrium and one ventricle
- **(C)** The capillaries inside the heart
- **(D)** The valves, not the chambers
- **(E)** The two ventricles
- **(F)** The veins entering the heart
- **(G)** The two atria (auricles)

36. A rectangular garden is 50 m by 30 m. A path 1 m wide runs all around the OUTSIDE of the garden. What is the area of the path? [Perimeter & Area]

- **(A)** 160 m^2
- **(B)** 1500 m^2
- **(C)** 84 m^2
- **(D)** 164 m^2
- **(E)** 1664 m^2
- **(F)** 320 m^2
- **(G)** 80 m^2

37. A tapeworm lives inside the human intestine and soaks up the food its host has already digested, giving nothing back. What kind of nutrition is this? [Nutrition in Animals]

- **(A)** Parasitic
- **(B)** Free-living holozoic
- **(C)** Insectivorous
- **(D)** Autotrophic
- **(E)** Photosynthetic
- **(F)** Saprophytic
- **(G)** Mutually helpful symbiosis

38. Write 72 as a product of prime numbers in power form. [Exponents & Powers]

- **(A)** 2 x 3^3
- **(B)** 2^4 x 3^2
- **(C)** 2^2 x 3^2
- **(D)** 2^3 x 3^2
- **(E)** 2^3 x 3^3
- **(F)** 2^3 x 3
- **(G)** 2^2 x 3^3

39. Human blood contains about 5,000,000 red blood cells in every cubic millimetre. Write this number in standard form. [Transportation in Animals & Plants]

- **(A)** 5 x 10^4
- **(B)** 5 x 10^6
- **(C)** 50 x 10^5
- **(D)** 5 x 10^5
- **(E)** 5 x 10^3
- **(F)** 0.5 x 10^7
- **(G)** 5 x 10^7

40. A square has an area of 81 cm^2. What is its perimeter? [Perimeter & Area]

- **(A)** 324 cm
- **(B)** 81 cm
- **(C)** 27 cm
- **(D)** 18 cm
- **(E)** 9 cm
- **(F)** 36 cm
- **(G)** 72 cm

41. Each gram of carbohydrate gives the body about 4 kcal of energy. A lunch supplies 75 g of carbohydrate. How much energy does this carbohydrate provide? [Nutrition in Animals]

- **(A)** 75 kcal
- **(B)** 300 kcal
- **(C)** 4 kcal
- **(D)** 79 kcal
- **(E)** 150 kcal
- **(F)** 1200 kcal
- **(G)** 37.5 kcal

42. A single cell divides into 2 cells every hour. Starting from 1 cell, how many cells are there after 6 hours? [Exponents & Powers]

- **(A)** 12
- **(B)** 128
- **(C)** 48
- **(D)** 32
- **(E)** 6
- **(F)** 64

43. A plant absorbs 500 mL of water from the soil. About 98% of it is lost by transpiration. How much water is left for the plant's own use? [Transportation in Animals & Plants]

- **(A)** 98 mL
- **(B)** 2 mL
- **(C)** 10 mL
- **(D)** 250 mL
- **(E)** 490 mL
- **(F)** 100 mL

44. In a rectangle, the length is twice the breadth, and the perimeter is 60 cm. What is the area of the rectangle? [Perimeter & Area]

- **(A)** 200 cm^2
- **(B)** 150 cm^2
- **(C)** 120 cm^2
- **(D)** 400 cm^2
- **(E)** 30 cm^2
- **(F)** 60 cm^2

45. Why do we chew our food thoroughly before swallowing instead of gulping it down in large pieces? [Nutrition in Animals]

- **(A)** To change the food's colour for the stomach
- **(B)** To add oxygen to the food for breathing
- **(C)** To break it into small bits and mix it with saliva so it digests faster
- **(D)** To turn the food directly into blood in the mouth
- **(E)** To kill every germ using the teeth
- **(F)** To cool the hot food before it reaches the stomach
- **(G)** To remove all the water from the food

46. Find the value of 10^3 + 10^2. [Exponents & Powers]

- **(A)** 1010
- **(B)** 110
- **(C)** 100000
- **(D)** 1100
- **(E)** 10^5
- **(F)** 2000
- **(G)** 1000

47. Why are the walls of capillaries made of only a single thin layer of cells? [Transportation in Animals & Plants]

- **(A)** So that germs cannot enter the blood
- **(B)** So that blood can flow very fast through them
- **(C)** So that they can stretch to hold air
- **(D)** So that they can pump blood like the heart
- **(E)** So that food and oxygen can pass easily between blood and body cells
- **(F)** So that they keep blood warm
- **(G)** So that they can store extra blood

48. A triangle has an area of 48 cm^2 and a base of 12 cm. What is its height? [Perimeter & Area]

- **(A)** 12 cm
- **(B)** 8 cm
- **(C)** 6 cm
- **(D)** 16 cm
- **(E)** 96 cm
- **(F)** 24 cm
- **(G)** 4 cm

49. The inner wall of the small intestine is folded into millions of tiny finger-like projections. What is their main job? [Nutrition in Animals]

- **(A)** To make the bile that digests fat
- **(B)** To push food backward into the stomach
- **(C)** To grind hard pieces of food into paste
- **(D)** To store undigested waste before it leaves
- **(E)** To kill germs with strong acid
- **(F)** To add water to dry food
- **(G)** To greatly increase the surface area for absorbing digested food

50. A spaceship covers 4 x 10^3 km on the first day and 5 x 10^2 km on the second day. Write the total distance in standard form. [Exponents & Powers]

- **(A)** 9 x 10^3 km
- **(B)** 9 x 10^5 km
- **(C)** 45 x 10^2 km
- **(D)** 4.5 x 10^4 km
- **(E)** 4.5 x 10^3 km
- **(F)** 4.05 x 10^3 km
- **(G)** 4.5 x 10^2 km

51. A person's pulse is 70 beats per minute at rest. After running, it rises by 60%. What is the new pulse rate? [Transportation in Animals & Plants]

- **(A)** 42
- **(B)** 182
- **(C)** 112
- **(D)** 130
- **(E)** 126
- **(F)** 60

52. A circle has a circumference of 66 cm. Taking pi = 22/7, what is its diameter? [Perimeter & Area]

- **(A)** 21 cm
- **(B)** 11 cm
- **(C)** 10.5 cm
- **(D)** 66 cm
- **(E)** 33 cm
- **(F)** 42 cm
- **(G)** 14 cm

53. A 250 g serving of spinach contains 4 mg of iron in every 100 g. How much iron is in the whole serving? [Nutrition in Animals]

- **(A)** 4 mg
- **(B)** 1.6 mg
- **(C)** 8 mg
- **(D)** 10 mg
- **(E)** 2.5 mg
- **(F)** 6.5 mg

54. Find the value of (2^2)^3. [Exponents & Powers]

- **(A)** 36
- **(B)** 512
- **(C)** 16
- **(D)** 256
- **(E)** 32
- **(F)** 64
- **(G)** 12

55. Each beat of the heart pushes out about 70 mL of blood. At 72 beats per minute, how much blood does the heart pump in one minute, in litres? [Transportation in Animals & Plants]

- **(A)** 142 mL
- **(B)** 0.504 L
- **(C)** 5.04 mL
- **(D)** 5.04 L
- **(E)** 50.4 L
- **(F)** 5040 L

56. A field has an area of 0.5 hectare. How many square metres is this? (1 hectare = 10000 m^2) [Perimeter & Area]

- **(A)** 5 m^2
- **(B)** 500 m^2
- **(C)** 25000 m^2
- **(D)** 50000 m^2
- **(E)** 50 m^2
- **(F)** 5000 m^2
- **(G)** 100 m^2

57. A meal has 50 g of protein. In the stomach, pepsin starts breaking down 30% of it, and the small intestine finishes the rest. How many grams of protein are digested in the small intestine? [Nutrition in Animals]

- **(A)** 45 g
- **(B)** 17.5 g
- **(C)** 35 g
- **(D)** 50 g
- **(E)** 15 g
- **(F)** 20 g
- **(G)** 30 g

58. If a = 2, find the value of a^3 x a^0 x a^2. [Exponents & Powers]

- **(A)** 12
- **(B)** 1
- **(C)** 32
- **(D)** 8
- **(E)** 64
- **(F)** 16
- **(G)** 2

59. When a person's kidneys fail and cannot clean the blood, which artificial process is used to remove the waste? [Transportation in Animals & Plants]

- **(A)** Bypass surgery
- **(B)** Skin grafting
- **(C)** Vaccination
- **(D)** Blood transfusion
- **(E)** Dialysis using a machine
- **(F)** Plastering the limb
- **(G)** A pacemaker

60. An L-shaped tile is made by taking an 8 cm by 8 cm square and cutting away a 3 cm by 4 cm rectangle from one corner. What is the area of the L-shape? [Perimeter & Area]

- **(A)** 56 cm^2
- **(B)** 64 cm^2
- **(C)** 12 cm^2
- **(D)** 76 cm^2
- **(E)** 52 cm^2
- **(F)** 48 cm^2
- **(G)** 60 cm^2

61. A child first grows 20 milk teeth. By adulthood the mouth has 32 teeth, of which 4 are wisdom teeth that appear last. Apart from the wisdom teeth, how many extra teeth are gained after the milk set? [Nutrition in Animals]

- **(A)** 4
- **(B)** 16
- **(C)** 8
- **(D)** 28
- **(E)** 20
- **(F)** 24
- **(G)** 12

62. Which one of these is NOT equal to 1? [Exponents & Powers]

- **(A)** 0^7
- **(B)** 5^0
- **(C)** (-3)^0
- **(D)** 1^9
- **(E)** 17^0
- **(F)** 8^0

63. During exercise a person sweats out 0.8 L of fluid. Sweat is about 99% water. Roughly how much water is lost as sweat? [Transportation in Animals & Plants]

- **(A)** 99 mL
- **(B)** 720 mL
- **(C)** 800 mL
- **(D)** 8 mL
- **(E)** 990 mL
- **(F)** 79.2 mL
- **(G)** 792 mL

64. A room floor is 6 m long and 5 m wide. Tiling it costs Rs 40 per square metre. What is the total cost of tiling? [Perimeter & Area]

- **(A)** Rs 880
- **(B)** Rs 1200
- **(C)** Rs 30
- **(D)** Rs 1500
- **(E)** Rs 600
- **(F)** Rs 60
- **(G)** Rs 440

65. Bread left damp for days grows a fungus that spreads thread-like growths over it and feeds on the dead bread, breaking it down. What mode of nutrition does this fungus show? [Nutrition in Animals]

- **(A)** Saprophytic
- **(B)** Symbiotic
- **(C)** Parasitic
- **(D)** Insectivorous
- **(E)** Photosynthetic
- **(F)** Autotrophic
- **(G)** Holozoic

66. Light travels about 3 x 10^5 km in one second. How far does it travel in 4 seconds, written in standard form? [Exponents & Powers]

- **(A)** 1.2 x 10^5 km
- **(B)** 3 x 10^9 km
- **(C)** 1.2 x 10^4 km
- **(D)** 12 x 10^5 km
- **(E)** 1.2 x 10^6 km
- **(F)** 12 x 10^6 km
- **(G)** 3 x 10^5 km

67. A heart beats 75 times per minute. About how many times does it beat in one full day? [Transportation in Animals & Plants]

- **(A)** 54000
- **(B)** 75
- **(C)** 10800
- **(D)** 1,08,000
- **(E)** 4500
- **(F)** 1800
- **(G)** 10,80,000

68. A semicircle has a radius of 7 cm. Taking pi = 22/7, what is the perimeter of the semicircle (curved part plus the straight diameter)? [Perimeter & Area]

- **(A)** 44 cm
- **(B)** 29 cm
- **(C)** 43 cm
- **(D)** 22 cm
- **(E)** 18 cm
- **(F)** 36 cm

69. A person's stomach empties a meal at a steady rate, taking 3 hours in all. If the meal weighs 600 g, how much of it is still in the stomach after 2 hours? [Nutrition in Animals]

- **(A)** 450 g
- **(B)** 400 g
- **(C)** 150 g
- **(D)** 100 g
- **(E)** 250 g
- **(F)** 200 g
- **(G)** 300 g

70. Which is the greatest: 3^4, 4^3, or 2^6? [Exponents & Powers]

- **(A)** 2^6
- **(B)** 6^2
- **(C)** 4^3 and 2^6 are both greatest
- **(D)** 4^3
- **(E)** All are equal
- **(F)** 3^4
- **(G)** 2^6 only

71. Which feature of veins stops blood from flowing backward as it returns to the heart? [Transportation in Animals & Plants]

- **(A)** Air bubbles that block the flow
- **(B)** A red pigment in their walls
- **(C)** Tiny hairs lining them
- **(D)** Valves inside the veins
- **(E)** Very thick muscular walls
- **(F)** A pumping action like the heart

72. A wire 88 cm long is bent to form a complete circle. Taking pi = 22/7, what is the radius of the circle? [Perimeter & Area]

- **(A)** 14 cm
- **(B)** 22 cm
- **(C)** 44 cm
- **(D)** 7 cm
- **(E)** 11 cm
- **(F)** 28 cm

73. A balanced plate is half vegetables, one-quarter grains, and the rest protein foods. If the plate holds 800 g of food, how many grams are protein foods? [Nutrition in Animals]

- **(A)** 200 g
- **(B)** 160 g
- **(C)** 100 g
- **(D)** 400 g
- **(E)** 250 g
- **(F)** 300 g
- **(G)** 600 g

74. Find the value of ((-3)^2)^2. [Exponents & Powers]

- **(A)** 6561
- **(B)** 12
- **(C)** 18
- **(D)** 81
- **(E)** -12
- **(F)** -81
- **(G)** -6

75. Of all the water a plant takes from the soil, only 1 part in 100 is actually used; the rest is transpired. If a plant transpires 297 mL in a day, how much water did it absorb? [Transportation in Animals & Plants]

- **(A)** 303 mL
- **(B)** 270 mL
- **(C)** 297 mL
- **(D)** 300 mL
- **(E)** 29700 mL
- **(F)** 99 mL

76. Square A has a side of 5 cm and square B has a side of 12 cm. Their two areas add up to the area of a single larger square. What is the side of that larger square? [Perimeter & Area]

- **(A)** 169 cm
- **(B)** 8.5 cm
- **(C)** 13 cm
- **(D)** 60 cm
- **(E)** 7 cm
- **(F)** 17 cm
- **(G)** 34 cm

77. Which row correctly traces the path of food through the human food canal from start to finish? [Nutrition in Animals]

- **(A)** Mouth, stomach, small intestine, oesophagus, anus, large intestine
- **(B)** Mouth, large intestine, stomach, small intestine, oesophagus, anus
- **(C)** Mouth, stomach, oesophagus, small intestine, large intestine, anus
- **(D)** Mouth, oesophagus, small intestine, stomach, large intestine, anus
- **(E)** Mouth, oesophagus, stomach, small intestine, large intestine, anus
- **(F)** Mouth, small intestine, stomach, oesophagus, large intestine, anus
- **(G)** Mouth, oesophagus, stomach, large intestine, small intestine, anus

78. Find the value of 9^2 - 3^4. [Exponents & Powers]

- **(A)** 81
- **(B)** 6
- **(C)** 0
- **(D)** 9
- **(E)** 18
- **(F)** -81

79. In a plant, the movement of food made in the leaves to other parts like roots and fruits is carried out by the phloem. What is this transport of food called? [Transportation in Animals & Plants]

- **(A)** Translocation
- **(B)** Pollination
- **(C)** Respiration
- **(D)** Photosynthesis
- **(E)** Germination
- **(F)** Filtration
- **(G)** Transpiration

80. A parallelogram has an area of 72 cm^2 and a height of 8 cm. What is the length of its base? [Perimeter & Area]

- **(A)** 64 cm
- **(B)** 576 cm
- **(C)** 36 cm
- **(D)** 18 cm
- **(E)** 4.5 cm
- **(F)** 9 cm

81. When kidneys and liver share the body's clean-up work, the liver makes bile and stores it in a small green sac before releasing it into the gut. What is this storage sac called? [Nutrition in Animals]

- **(A)** Gall bladder
- **(B)** Pancreas
- **(C)** Food vacuole
- **(D)** Spleen
- **(E)** Appendix
- **(F)** Urinary bladder
- **(G)** Salivary gland

82. There are 86,400 seconds in a day. Write this number in standard form. [Exponents & Powers]

- **(A)** 0.864 x 10^5
- **(B)** 8.64 x 10^4
- **(C)** 8.64 x 10^5
- **(D)** 8.64 x 10^3
- **(E)** 864 x 10^2
- **(F)** 8.64 x 10^2

83. Through which structures does most of a plant's water loss (transpiration) take place? [Transportation in Animals & Plants]

- **(A)** The colourful petals of flowers
- **(B)** The waxy upper surface of the leaf
- **(C)** The seeds inside the fruit
- **(D)** The thick bark of the stem
- **(E)** The tip of the main root
- **(F)** The woody xylem tubes only
- **(G)** The stomata, mostly on the underside of leaves

84. A rectangle is 10 cm long and 8 cm wide. If the length is increased by 50% while the width stays the same, what is the new area? [Perimeter & Area]

- **(A)** 240 cm^2
- **(B)** 40 cm^2
- **(C)** 160 cm^2
- **(D)** 120 cm^2
- **(E)** 100 cm^2
- **(F)** 60 cm^2
- **(G)** 80 cm^2

85. Grass is rich in cellulose, which humans cannot digest. How do cattle manage to get food value from grass that we cannot? [Nutrition in Animals]

- **(A)** Cellulose passes straight into their blood unchanged
- **(B)** They have a special acid in saliva that melts cellulose
- **(C)** Their teeth grind cellulose into sugar directly
- **(D)** They swallow stones that crush the cellulose
- **(E)** Their liver turns cellulose into bile
- **(F)** Helpful microbes in their rumen break the cellulose down for them
- **(G)** They photosynthesise the grass like a plant

86. A square has each side equal to 2^3 cm. What is its area? [Exponents & Powers]

- **(A)** 64 cm^2
- **(B)** 16 cm^2
- **(C)** 32 cm^2
- **(D)** 2^9 cm^2
- **(E)** 36 cm^2
- **(F)** 12 cm^2
- **(G)** 48 cm^2

87. Why is the circulation of blood in humans called a 'double circulation'? [Transportation in Animals & Plants]

- **(A)** Because the heart beats in two halves a second apart
- **(B)** Because blood passes through the heart twice for one complete round of the body
- **(C)** Because blood flows in two directions in the same vessel
- **(D)** Because blood is pumped only twice a day
- **(E)** Because there are two kinds of blood that never mix outside
- **(F)** Because there are two hearts in the body

88. A rectangle is 20 cm by 14 cm. A triangle is drawn inside it with a base of 20 cm and a height of 14 cm. What is the area of the triangle? [Perimeter & Area]

- **(A)** 68 cm^2
- **(B)** 210 cm^2
- **(C)** 280 cm^2
- **(D)** 34 cm^2
- **(E)** 160 cm^2
- **(F)** 140 cm^2

89. A doctor tells a patient that one part of the digestive system mainly absorbs water from leftover food and forms semi-solid waste. Which part is this? [Nutrition in Animals]

- **(A)** Oesophagus
- **(B)** Mouth
- **(C)** Gall bladder
- **(D)** Large intestine
- **(E)** Liver
- **(F)** Stomach

90. Simplify (5^2 x 5^4) / 5^3 and give its value. [Exponents & Powers]

- **(A)** 3125
- **(B)** 15
- **(C)** 25
- **(D)** 5
- **(E)** 625
- **(F)** 125

91. Water and dissolved minerals enter a plant mainly through which part? [Transportation in Animals & Plants]

- **(A)** The woody bark
- **(B)** The tip of the shoot
- **(C)** The fruit skin
- **(D)** The leaf stomata
- **(E)** The flower petals
- **(F)** The root hairs

92. A circular pond has a radius of 14 m. Taking pi = 22/7, what is the area of the pond? [Perimeter & Area]

- **(A)** 616 m^2
- **(B)** 1232 m^2
- **(C)** 154 m^2
- **(D)** 308 m^2
- **(E)** 88 m^2
- **(F)** 196 m^2
- **(G)** 44 m^2

93. Hydra is a tiny water animal that uses stinging tentacles to catch small prey, then takes the prey inside its body to digest it. What type of nutrition is Hydra showing? [Nutrition in Animals]

- **(A)** Saprophytic nutrition
- **(B)** Photosynthetic nutrition
- **(C)** Autotrophic nutrition
- **(D)** Insectivorous plant nutrition
- **(E)** Parasitic nutrition
- **(F)** Holozoic nutrition
- **(G)** Mutual symbiosis

94. Find the value of 7^5 / 7^3. [Exponents & Powers]

- **(A)** 14
- **(B)** 7^8
- **(C)** 16
- **(D)** 49
- **(E)** 7
- **(F)** 2401
- **(G)** 343

95. The clear, slightly yellow fluid that leaks out of blood capillaries into the tissues, helps fight germs, and drains back through its own vessels is called what? [Transportation in Animals & Plants]

- **(A)** Bile
- **(B)** Lymph
- **(C)** Plasma
- **(D)** Saliva
- **(E)** Mucus
- **(F)** Urine

96. A square plot has a side of 35 m. A wall already stands along one side, so fencing is needed on only the other three sides. Fencing costs Rs 12 per metre. What is the cost? [Perimeter & Area]

- **(A)** Rs 1260
- **(B)** Rs 840
- **(C)** Rs 1680
- **(D)** Rs 420
- **(E)** Rs 1225
- **(F)** Rs 105
- **(G)** Rs 630

97. In an experiment, iodine is dropped on a sample. Iodine turns blue-black only if starch is present. After bread is fully chewed with saliva and tested, the blue-black colour is much weaker. What does this show? [Nutrition in Animals]

- **(A)** Chewing turned starch into protein
- **(B)** Saliva has digested much of the starch in the bread
- **(C)** Saliva turned the starch into fat
- **(D)** The bread never had any starch
- **(E)** Saliva added more starch to the bread
- **(F)** Iodine destroyed the saliva

98. Find the value of 6^3 - 6^2. [Exponents & Powers]

- **(A)** 6
- **(B)** 180
- **(C)** 30
- **(D)** 216
- **(E)** 36
- **(F)** 252
- **(G)** 1080

99. A patient's blood pressure is much higher in the arteries than in the veins. Which statement best explains why? [Transportation in Animals & Plants]

- **(A)** Arteries carry more blood cells, raising the pressure
- **(B)** Veins are nearer the heart, so they get more pressure
- **(C)** Arteries are colder, which raises the pressure
- **(D)** Arteries receive blood pumped straight from the heart, so the pressure is high
- **(E)** Veins are wider, so the pressure must be higher in them
- **(F)** Arteries are filled only with air at high pressure
- **(G)** Veins have thicker walls, so they hold higher pressure

100. Two identical rectangles, each 6 m by 4 m, are placed side by side and joined along a 4 m edge to make one bigger rectangle. What is the perimeter of the bigger rectangle? [Perimeter & Area]

- **(A)** 48 m
- **(B)** 40 m
- **(C)** 20 m
- **(D)** 36 m
- **(E)** 24 m
- **(F)** 32 m